



WOLLO University

Department of software engineering

Exit multiple choice questions for

Software project management

Prepared by:

@Neyatibeb

KOMBOLCHA, Ethiopia

2015 E.C

1. Which of the following is not a project management goal?

- A. Avoiding customer complaints
- B. Keeping overall costs within budget
- C. Delivering the software to the customer at the agreed time
- D. Maintaining a happy and well-functioning development team

View Answer

Answer : A

Explanation: Avoiding customer complaints is not a project management goal. The project management goals are: Keeping overall costs within budget, Maintaining a happy and well-functioning development team and Delivering the software to the customer at the agreed time.

2. What is the first step in project planning?

- A. Determine the budget
- B. Determine project constraints
- C. Inspect the deliverables
- D. Establish the objectives and scope

View Answer

Answer : D

Explanation: The first step in project planning is to establish the objectives and scope. So option D is correct.

3. A project is anything which is

- A. Not implicitly expressed
- B. Not a physical objective
- C. Implicitly expressed
- D. Social acceptability

View Answer

Answer : C

Explanation: A project is anything which is implicitly expressed.

4. Process framework activities are populated with

- A. Milestones
- B. QA Points
- C. Work Products
- D. All of the above

View Answer

Answer : D

Explanation: Process framework activities are populated with Milestones, QA Points, Work Products. So All of the above option is correct.

5. Which of the following is not an effective software project management focus?

- A. People
- B. Popularity
- C. Product
- D. Process

View Answer

Answer : B

Explanation: Popularity is not an effective software project management focus. The four P's of effective software project management are: People, Product, Process and Project.

6. The reason for software bugs and failures is due to

- A. Software developers
- B. Software companies
- C. Both (A) and (B)
- D. None of the above

View Answer

Answer : C

Explanation: Bugs from software developers side is no new thing and software companies are responsible for making policies and providing working atmosphere for the software development, so companies become a part of software development process and due to this, Software developers as well as Software companies both can be a reason for software bugs and failures.

7. According to the Project Management Institute (PMI), project management is defined as “the application of knowledge, _____, _____, and techniques to project activities to meet the project requirements”.

- A. Skills, Tools
- B. Skills, Analysis
- C. Tools, Analysis
- D. Analysis, Theories
- E. Skills, Theories

View Answer

Answer : A

Explanation: According to the Project Management Institute (PMI), project management is defined as “the application of knowledge, Skills, Tools, and techniques to project activities to meet the project requirements”.

8. During project estimation, project manager estimates following

- A. Project duration
- B. Project effort
- C. Project cost
- D. All of the above

View Answer

Answer : D

Explanation: During project estimation, project manager estimates following: Project duration, Project effort, Project cost. So All of the above option is correct.

9. Project managers have to assess the risks that may affect a project

- A. True
- B. False

View Answer

Answer : B

Explanation: Risk management involves anticipating risks that might affect the project schedule or the quality of the software being developed, and then taking action to avoid these risks and project managers don't have to assess the risks that may affect a project.

10. Different activity of a project management is

- A. Project planning
- B. Project monitoring
- C. Project control
- D. All of the above

View Answer

Answer : D

11. Cultural implications of the project are examined by

- A. Cultural feasibility
- B. Economic feasibility
- C. Technical feasibility
- D. Marketing feasibility

View Answer

Answer : A

Explanation: Cultural implications of the project are examined by Cultural feasibility.

12. Effective software project management focuses on

- A. People, Performance, Payoff, Product
- B. People, Product, Process, Project
- C. People, Process, Payoff, Product
- D. People, Product, Performance, Process

View Answer

Answer : B

Explanation: Effective software project management focuses on four P's: People, Product, Process, Project.

13. Which of these are not among the eight principles followed by Software Engineering Code of Ethics and Professional Practice ?

- A. Public
- B. Product
- C. Profession
- D. Environment

View Answer

Answer : D

Explanation: Environment is not among the eight principles followed by Software Engineering Code of Ethics and Professional Practice, rest all are clauses for software ethics.

14. The _____ is not an approach to software cost estimation?

- A. Analytical
- B. Heuristic
- C. Critical
- D. Empirical

View Answer

Answer : C

Explanation: The critical approach is not an approach to software cost estimation.

15. Which of these does not account for software failure?

- A. Increasing demand
- B. Less reliable and expensive
- C. Increasing supply
- D. Low expectation

View Answer

Answer : C

Explanation: Increasing supply does not account for software failure because it will lead to more production but not failure.

PHP MCQ Questions And Answers

16. Which of the following is not considered as a risk in project management?

- A. Testing

- B. Specification delays
- C. Product competition
- D. Staff turnover

View Answer

Answer : A

Explanation: Testing is not considered as a risk in project management because it is a part of project.

17. What are attributes of good software ?

- A. Software maintainability
- B. Software maintainability & functionality
- C. Software functionality
- D. Software development

View Answer

Answer : B

Explanation: Software maintainability & functionality are the attributes of a good software because a good software should deliver the required functionality and maintainability.

18. Risk management is one of the most important jobs for a

- A. Investor
- B. Client
- C. Production team
- D. Project manager

View Answer

Answer : D

Explanation: Risk management is one of the most important jobs for a Project manager.

19. Capacity planning

- A. Increase cost
- B. Save time
- C. Increases waste
- D. Increase production

View Answer

Answer : B

Explanation: Capacity planning Saves time.

20. Which of the following is not considered a stakeholder in the software process?

- A. Project Managers
- B. End-Users
- C. Customers
- D. Sales People

View Answer

Answer : D

Explanation: Sales People is not considered a stakeholder in the software process. However, Customers, End-Users, Project Managers are considered as a stakeholder in the software process.

21. Which paradigm structures a team loosely and depends on individual initiatives of the team members?

- A. Synchronous paradigm
- B. Random paradigm
- C. Open paradigm
- D. Closed paradigm

View Answer

Answer : A

Explanation: Synchronous paradigm structures a team loosely and depends on individual initiatives of the team members. So option A is correct.

22. Which of these is true ?

- A. Generic products are produced by organization and sold to open market
- B. Generic products and customized products are types of software products
- C. Customized products are commissioned by particular customer
- D. All of the above

View Answer

Answer : D

Explanation: Generic products are produced by organization and sold to open market, generic products and customized products are types of software products and customized products are commissioned by particular customer, all of them are correct.

23. The process each manager follows during the life of a project is known as

- A. Manager life cycle
- B. Project Management
- C. Project Management Life Cycle
- D. None of the above

View Answer

Answer : C

Explanation: The process each manager follows during the life of a project is known as Project Management Life Cycle.

24. The model serves as the basis for the creation of _____ for the Software.

- A. Specifications
- B. Testing
- C. Maintenance
- D. Design

View Answer

Answer : A

Explanation: The model serves as the basis for the creation of Specifications for the Software.

25. Which of the following activity is undertaken immediately after feasibility study and before the requirement analysis and specification phase?

- A. Project Monitoring
- B. Project Planning
- C. Project Control
- D. Project Scheduling

View Answer

Answer : B

Explanation: Project Planning is undertaken immediately after feasibility study and before the requirement analysis and specification phase.

Software Engineering MCQ Questions And Answers

26. A 66.6% risk is considered as

- A. High
- B. Low
- C. Very Low
- D. Moderate

View Answer

Answer : A

Explanation: The probability of the risk might be assessed as very low (<10%), low (10–25%), moderate (25–50%), high (50–75%), or very high (>75%). So in this case, the risk is considered high because it comes in the range (50-75%).

27. Small agile teams have no place in modern software development

- A. True
- B. False

View Answer

Answer : B

Explanation: False statement.

28. Which one of the following is not a software process quality?

- A. Productivity
- B. Timeliness
- C. Visibility
- D. Portability

View Answer

Answer : D

Explanation: Portability is not a software process quality but it is a software product quality which means software can run on different hardware platforms or software environments.

29. Which of the following is/are main parameters that you should use when computing the costs of a software development project?

- A. Travel and training costs
- B. Hardware and software costs
- C. Effort costs (the costs of paying software engineers and managers)
- D. All of the above

View Answer

Answer : D

Explanation: All of the above like Travel and training costs, Hardware and software costs, and Effort costs (the costs of paying software engineers and managers) are main parameters that you should use when computing the costs of a software development project.

30. When can selected common process framework activities be omitted during process decomposition?

- A. When the project is extremely small in size
- B. Never the activities are invariant
- C. Anytime the software is mission critical
- D. Rapid prototyping does not require their use

View Answer

Answer : B

Explanation: Never, the activities are invariant.

1. Which of the following is incorrect activity for the configuration management of a software system?

1. system management
2. change management
3. internship management

4. version management

Answer: internship management

2. What is one or more software configuration items that have been formally reviewed and agreed upon and serve as a basis for further development?

1. cm
2. cumulative changes
3. baseline
4. change control

Answer: baseline

3. Which of the following is not a main phase in Configuration Management (CM) Process?

1. cm audits
2. executing the cm process
3. cm planning
4. None

Answer: None

4. Which of the following activities of a Generic Process framework provides a feedback report?

1. planning
2. communication
3. modeling & construction
4. deployment

Answer: deployment

5. Which of the following is not project management goal?

1. maintaining a happy and well-functioning development team
2. delivering the software to the customer at the agreed time
3. keeping overall costs within budget
4. avoiding customer complaints

Answer: avoiding customer complaints

6. Identify the sub-process of process improvement _____

1. de-processification
2. process analysis
3. process introduction
4. process distribution

Answer: process analysis

7. Purpose of process is to deliver software _____

1. with acceptable quality
2. in time
3. that is cost efficient
4. both in time & with acceptable quality

Answer: both in time & with acceptable quality

8. Which of these truly defines Software design?

1. software design satisfies client needs and desires
2. software design specifies nature and composition of software product
3. software design is an activity subjected to constraints
4. All of the above

Answer: All of the above

9. Which one of the following is not an application of embedded software product?

1. pattern recognition game playing
2. keypad control of a security system
3. digital function of dashboard display in a car
4. None

Answer: pattern recognition game playing

10. Which of the following is not considered as a risk in project management?

1. testing
2. product competition
3. specification delays
4. staff turnover

Answer: testing

11. Which of these describes step wise refinement?

1. it is a top down approach
2. stepwise refinement follows its existence from 1971
3. nicklaus with described the first software engineering method as step wise refinement
4. All of above

Answer: All of above

12. Which one of the following is not a software process quality?

1. productivity
2. portability
3. timeliness
4. visibility

Answer: portability

13. The fundamental notions of software engineering does not account for ?

1. software security
2. software processes
3. software reuse
4. software validation

Answer: software validation

14. Which of these does not affect different types of software as a whole?

1. flexibility
2. business and social change
3. heterogeneity
4. security

Answer: flexibility

15. The process each manager follows during the life of a project is known as _____

1. project management life cycle
2. manager life cycle
3. project management
4. All of the above

Answer: project management life cycle

16. What is the solution for Structural design?

1. specification model is structure chart showing procedure calling hierarchy and flow of data in and out of procedures
2. the specification model following data flow diagram
3. procedures represented as bubbles
4. emphasizing procedural decomposition

Answer: specification model is structure chart showing procedure calling hierarchy and flow of data in and out of procedures

17. What are attributes of good software ?

1. software functionality
2. software maintainability
3. software development
4. software maintainability & functionality

Answer: software maintainability & functionality

18. Which of these does not account for software failure ?

1. low expectation
2. increasing supply
3. increasing demand
4. less reliable and expensive

Answer: increasing supply

19. Which of the following is an incorrect method for structural design?

1. designing object oriented systems
2. handling of larger and more complex products
3. transition of problem models to solution models
4. more procedural approach

Answer: handling of larger and more complex products

20. A 66.6% risk is considered as _____

1. very low
2. moderate
3. low
4. high

Answer: high

21. Which of these are not among the eight principles followed by Software Engineering Code of Ethics and Professional Practice ?

1. profession
2. public
3. product
4. environment

Answer: environment

22. What is followed by the design task?

1. following design task heuristics
2. checking model's completeness
3. choosing specific classes, operations
4. All of above

Answer: All of above

23. The reason for software bugs and failures is due to _____

1. software companies
2. software developers
3. both software companies and developer
4. all of the mentioned

Answer: both software companies and developer

24. Efficiency in a software product does not include _____

1. A memory utilization
2. licensing
3. responsiveness
4. processing time

Answer: licensing

25. Which of the following is/are main parameters that you should use when computing the costs of a software development project?

1. effort costs (the costs of paying software engineers and managers)
2. hardware and software costs
3. travel and training costs
4. All of the above

Answer: All of the above

26. Identify an ethical dilemma from the situations mentioned below

1. refusing to undertake a project
2. your employer releases a safety-critical system without finishing the testing of the system
3. agreement in principle with the policies of senior management
4. all of the mentioned

Answer: your employer releases a safety-critical system without finishing the testing of the system

27. Which of this analysis are not acceptable?

1. object oriented design are given more preference than structural design
2. object oriented design always dominates structural design
3. object oriented design is a far better approach compared to structural design
4. object oriented uses more specific notation

Answer: object oriented design always dominates structural design

28. Explain what is meant by **PRODUCT** with reference to one of the eight principles as per the ACM/IEEE Code of Ethics ?

1. software engineers shall ensure that their products and related modifications satisfy the client
2. software engineers shall ensure that their products and related modifications meet the highest professional standards possible
3. the product should be easy to use
4. it means that the product designed /created should be easily available

Answer: software engineers shall ensure that their products and related modifications meet the highest professional standards possible

29. Which of these does not represent object oriented design?

1. central model represents class diagrams that show the classes comprising a program and their relationships to one another
2. programs are thought of collection of objects
3. it follows regular procedural decomposition in favor of class and object decomposition
4. object-oriented methods incorporates structural methods

Answer: it follows regular procedural decomposition in favor of class and object decomposition

30. Quality planning is the process of developing a quality plan for _____

1. team
2. customers
3. project
4. project manager

Answer: project

Applicant constant touch with our **dailyrecruitment** site regularly to get latest GK Quiz, upcoming exam notification, central government/state government jobs etc.

1. The aim of an organizational structure is to facilitate cooperation towards a common goal. (True / False)

Ans. True

2. Cutting out unnecessary requirements in the software project is called ____.

Ans. Requirements Scrubbing

3. ____ team organization attempts to combine the benefits of centralized and decentralized control while minimizing or avoiding their disadvantages.

Ans. Mixed-Control

4. Decentralized control is best when communication among engineers is necessary for achieving a good solution. (True / False)

Ans. True

5. ____ is best when the speed of development is the most important goal and the problem is well understood.

Ans. Centralized-Control

6. ____ refers to the long-term management of intractable conflicts.

Ans. Conflict Management

7. The risk associated with the degree of uncertainty that the product will meet its requirements and be fit for its intended use is a ____.

- a) Cost risk
- b) Performance risk
- c) Support risk
- d) Schedule risk

Ans. b) Performance risk

8. SEI-CMM stands for ____.

Ans. Software Engineering Institute – Capability Maturity Model

9. ANSI stands for ____.

Ans. American National Standards Institute

10. Software quality assurance (SQA) is an umbrella activity that is applied throughout the lifecycle of the project across all phases. (True / False)

Ans. True

11. The term CASE is applied to software product development that uses extensive software engineering principles and these processes are implemented either partly or majorly through the supporting software. (True / False)

Ans. True

12. CASE stands for ____.

Ans. Computer-Aided Software Engineering

13. The ____ is the heart of a CASE system.

a) Repository

b) Dictionary

c) Design

d) Coding

Ans. a) Repository

14. Business process engineering tools provide the basic model of the information which acts as the meta-data from which the specific information can be derived. (True / False)

Ans. True

15. ____ model the processes by representing the key elements of the process.

Ans. Process modeling tools

16. Screen painter is an example of ____ tools.

a) Programming

b) Prototyping

c) Design

d) Analysis

Ans. b) Prototyping

17. The term integration implies both combination and closure. (True /False)

Ans. True

18. SCI stands for ____.

Ans. Software Configuration Items

19. ____ control the behavior of tools within the environment.

Ans. Tools management services

20. ____ is a database that acts as the center for both accumulation and storage of software engineering information.

Ans. CASE Repository

21. Software testing is the process of running through the application or a software product with the intention of uncovering the errors. (True / False)

Ans. True

22. ____ to ensure that the software will function well when the number of users is increased on the system.

Ans. Scalability Testing

23. Software testing typically represents ____ of a software development budget.

a) 40%

b) 20%

c) 70%

d) 90%

Ans. a) 40%

24. Black Box Testing tests the internal structure of the system. (True / False)

Ans. False

25. ____ method leads to a selection of test cases that exercise boundary values.

Ans. Boundary Value Analysis

26. ____ technique provides a logical representation of various possible operational scenarios of the application being tested.

- a) Boundary Value Analysis
 - b) Cause-effect graphing
 - c) Equivalence Partitioning
 - d) Basis Path Testing
- Ans. b) Cause-effect graphing

27. White box testing is a test case design method that uses the control structure of the procedural design to derive test cases. (True / False)

Ans. True

28. ____ is a software metric that provides a quantitative measure of the logical complexity of a program.

Ans. Cyclomatic complexity

29. Which one of the following is a Software Testing tool?
- a) Rational Rose
 - b) Oracle
 - c) Win Runner
 - d) Linux
- Ans. c) Win Runner

30. ____ is an SQA activity that assures standards and procedures defined for the project are followed properly and follows the compliance as stated.

Ans. Product evaluation

31. Software maintenance is the modification of a software product after delivery to correct errors, improve performance, or adapt to new requirements. (True / False)

Ans. True

32. ____ Allow legacy applications to adapt to changing requirements in quick time.

Ans. Reengineering

33. In a recent study, the top 10 most expensive software errors were ____ errors.
- a) Maintenance
 - b) Analysis
 - c) Design
 - d) Coding
- Ans. a) Maintenance

34. Legacy application of the organization holds a lot of business-critical details and represents the enterprise architecture of the corporation. (True / False)

Ans. True

35. Around 80% of the coding of the legacy applications is done using ____.

Ans. COBOL

36. ____ is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service, and speed.
- a) Software Engineering
 - b) Business Process Reengineering
 - c) Software Reengineering
 - d) Business Process Outsourcing
- Ans. b) Business Process Reengineering

37. Reengineering of information systems is an activity that will absorb information technology resources for a short time. (True / False)

Ans. False

38. There are ____ phases in the life-cycle of the software reengineering process model.

Ans. Six

39. ____ for software is the process of analyzing a program in an effort to create a representation of the program at a higher level of abstraction than source code.

- a) Software Engineering
- b) Business Process Reengineering
- c) Reverse engineering
- d) Business Process Outsourcing

Ans. c) Reverse Engineering

40. Reengineering tools are platform-independent and can work on any platform. (True / False)

Ans. False

41. I-CASE stands for ____.

Ans. Integrated Computer-Aided Software Engineering

42. The data obtained during the closure analysis are used to populate the ____.

Ans. Process Database (PDB)

43. The final project closure analysis report is submitted to the ____ of the project.

- a) Business Manager
- b) Project Manager
- c) System Manager
- d) Database Manager

Ans. a) Business Manager

44. At Infosys, many projects use the top-down method for estimation. (True / False)

Ans. False

45. For normalization purposes, the productivity of a project is measured in terms of ____ per person-month.

Ans. Function Points (FP)

46. Lack of support from the database architect and database administrator of the customer is considered a risk. (True / False)

Ans. True

47. RUP stands for ____.

Ans. Rational Unified Process (RUP)

48. ____ is an SQA activity that ensures that appropriate steps to carry out the process are being followed.

Ans. Process monitoring

49. Software Reliability is measured in Mean-Time-To-Failure (MTTF). (True / False)

Ans. True

50. ____ is defined as the ease of finding and correcting errors in the software.

Ans. Software maintainability

1) What is the first step in the software development lifecycle?

- a. System Design
- b. Coding
- c. System Testing
- d. Preliminary Investigation and Analysis

Answer: d) Preliminary Investigation and Analysis

Explanation: The first phase of the software development life cycle is the requirement analysis, in which the software engineer analyzes the software requirement based on the user's outlook.

2) What does the study of an existing system refer to?

- a. Details of DFD
- b. Feasibility Study
- c. System Analysis
- d. System Planning

Answer: c) System Analysis

Explanation: The study of an existing system refers to system analysis.

3) Which of the following is involved in the system planning and designing phase of the Software Development Life Cycle (SDLC)?

- a. Sizing
- b. Parallel run
- c. Specification freeze
- d. All of the above

Answer: d) All of the above

Explanation: None.

4) What does RAD stand for?

- a. Rapid Application Document
- b. Rapid Application Development
- c. Relative Application Development
- d. None of the above

Answer: b) Rapid Application Development

Explanation: RAD stands for Rapid Application Development is categorized as an agile development method, which is meant to accomplish a quick turnaround and high-end outcomes.

5) Which of the following prototypes does not associated with Prototyping Model?

- a. Domain Prototype
- b. Vertical Prototype
- c. Horizontal Prototype

- d. Diagonal Prototype

Answer: d) Diagonal Prototype

Explanation: There is nothing as such called Diagonal Prototype; however, the rest other options do make sense.

6) The major drawback of RAD model is _____.

- a. It requires highly skilled developers/designers.
- b. It necessitates customer feedbacks.
- c. It increases the component reusability.
- d. Both (a) & (c)

Answer: d) Both (a) & (c)

Explanation: The client may create an unrealistic product vision, which may result in a team over or under-develop functionality. Besides, it is not easy to hire highly specialized & skilled developers/designers.

7) Which of the following does not relate to Evolutionary Process Model?

- a. Incremental Model
- b. Concurrent Development Model
- c. WINWIN Spiral Model
- d. All of the above

Answer: d) All of the above

Explanation: The evolutionary process model is designed for rapid software development, which starts from scratch with raw specifications and modifies as per the user requirement.

8) What is the major drawback of the Spiral Model?

- a. Higher amount of risk analysis
- b. Doesn't work well for smaller projects
- c. Additional functionalities are added later on
- d. Strong approval and documentation control

Answer: b) Doesn't work well for smaller projects

Explanation: A spiral model is an incremental approach, which is formed as a combination of the waterfall model and prototyping model. The major drawbacks of the Spiral model are as follows:

- Expensive
- Doesn't work well for smaller projects

- Risk analysis requires highly skilled experts.

9) Model selection is based on _____.

- a. Requirements
- b. Development team & users
- c. Project type & associated risk
- d. All of the above

Answer: d) All of the above

Explanation: Model selection is based on requirements, a team of developers, users as well as the risk involved in developing a project.

10) Which of the following option is correct?

- a. The prototyping model facilitates the reusability of components.
- b. RAD Model facilitates reusability of components
- c. Both RAD & Prototyping Model facilitates reusability of components
- d. None

Answer: c) Both RAD & Prototyping Model facilitates reusability of components

Explanation: None.

11) Which of the following models doesn't necessitate defining requirements at the earliest in the lifecycle?

- a. RAD & Waterfall
- b. Prototyping & Waterfall
- c. Spiral & Prototyping
- d. Spiral & RAD

Answer: c) Spiral & Prototyping

Explanation: In the Prototyping model, the first phase is the requirement analysis phase, which involves Brainstorming, QFD, and FAST, whereas the Spiral model encompasses customer communication activities such as defining objectives.

12) When the user participation isn't involved, which of the following models will not result in the desired output?

- a. Prototyping & Waterfall
- b. Prototyping & RAD
- c. Prototyping & Spiral

- d. RAD & Spiral

Answer: b) Prototyping & RAD

Explanation: The Prototyping model necessitates user participation as when a prototype is created or designed, whereas, in the case of RAD, user participation is required in all four phases.

13) Which of the following model will be preferred by a company that is planning to deploy an advanced version of the existing software in the market?

- a. Spiral
- b. Iterative Enhancement
- c. RAD
- d. Both (b) and (c)

Answer: d) Both (b) and (c)

Explanation: None.

14) Arrange the following activities for making a software product by utilizing 4GT.

I. Design strategy

II. Transformation into product

III. Implementation

IV. Requirement gathering

- a. 4, 1, 3, 2
- b. 4, 3, 2, 1
- c. 1, 2, 3, 4
- d. 1, 4, 2, 3

Answer: a) 4, 1, 3, 2

Explanation: The sequence of activities mentioned in option c represents the Fourth Generation Techniques(4GT) Model.

15) Which of the following is an example of Black Box and Functional Processing?

- a. First Generation Language
- b. Second Generation Language
- c. Third Generation Language
- d. Fourth Generation Language

Answer: d) Fourth Generation Language

Explanation: Functional processing or testing is commonly known as black-box testing, where the existing contents are almost unknown, so anything could be demonstrated as a black box. Hence the functionality of a black box can be defined in terms of inputs and outputs.

16) _____ is identified as fourth generation language.

- a. Unix shell
- b. C++
- c. COBOL
- d. FORTRAN

Answer: a) Unix shell

Explanation: Fourth-generation language is an advancement of third-generation programming language.

17) The productivity of a software engineer can be reduced by using a 4GT.

- a. True
- b. False

Answer: b) False

Explanation: The fourth-generation programming languages are more programmer-friendly and helpful in augmenting the program's efficiency as it uses English words and phrases. It increases the programmer's productivity by enabling them to engage in the software development life cycle.

18) Which of the following is the main advantage of deploying a 4GT model for producing small-scale products, programs, and applications?

- a. The productivity of software engineers is improved.
- b. The time required for developing software is reduced.
- c. CASE tools and code generators help the 4GT model by providing a credible solution to their problems.
- d. None of the above.

Answer: b) Time required for developing software is reduced.

Explanation: Since automated coding is done using CASE tools & code generators, proponents claim a dramatic reduction in software development time.

19) Which of the following model has a major downfall to a software development life cycle in terms of the coding phase?

- a. 4GT Model
- b. Waterfall Model
- c. RAD Model
- d. Spiral Model

Answer: a) 4GT Model

Explanation: Much more expertise is needed in the 4GT model for analyzing, designing, and testing activities as it eliminates the coding phase.

20) Which of the following falls under the category of software products?

- a. Firmware, CAD
- b. Embedded, CAM
- c. Customized, Generic
- d. CAD, Embedded

Answer: c) Customized, Generic

Explanation: Customized and Generic are two kinds of software products.

21) Software maintenance costs are expensive in contrast to software development.

- a. True
- b. False

Answer: a) True

Explanation: For systems with long life, software maintenance cost is higher than that of the software development cost.

22) Which of the following activities of the generic process framework delivers a feedback report?

- a. Deployment
- b. Planning
- c. Modeling
- d. Construction

Answer: a) Deployment

Explanation: The deployment phase is the last phase of the software development life cycle in which the software product is delivered to its end-user, who further assesses its performance and revert back with the feedback if anything is required or missing as per the formulated evaluation.

23) Which of the following refers to internal software equality?

- a. Scalability
- b. Reusability
- c. Reliability
- d. Usability

Answer: b) Reusability

Explanation: None.

24) RUP is abbreviated as _____, invented by a division of _____.

- a. Rational Unified Process, IBM
- b. Rational Unified Program, IBM
- c. Rational Unified Process, Infosys
- d. Rational Unified Program, Infosys

Answer: a) Rational Unified Process, IBM

Explanation: None.

25) The RUP can be defined based on three main perspectives, i.e., static, dynamic & practice. What is the purpose of static perspective?

- a. It recommends some good practices that be included while carrying out the process.
- b. It displays all the enacted process activities.
- c. It portrays the phases that the model has entered over a specific time.
- d. All of the above.

Answer: b) It displays all the enacted process activities.

Explanation: None.

26) Which one of the following activities is not recommended for software processes in software engineering?

- a. Software Evolution
- b. Software Verification
- c. Software Testing & Validation
- d. Software designing

Answer: b) Software Verification

Explanation: Software verification is mainly considered for implementing and testing activities.

27) Arrange the following activities to form a general software engineering process model.

I. Manufacture

II. Maintain

III. Test

IV. Install

V. Design

VI. Specification

- a. 6, 5, 1, 3, 4, 2
- b. 1, 2, 4, 3, 6, 5
- c. 6, 1, 4, 2, 3, 5
- d. 1, 6, 5, 2, 3, 4

Answer: a) 6, 5, 1, 3, 4, 2

Explanation: None.

28) The agile software development model is built based on _____.

- a. Linear Development
- b. Incremental Development
- c. Iterative Development
- d. Both Incremental and Iterative Development

Answer: d) Both Incremental and Iterative Development

Explanation: In agile software development, the software is usually developed in increments with the customer specifying necessary requirements in each increment. Here the main aim of a developer is to meet customer satisfaction at the earliest by providing a valuable software product. It is known as iterative because it chooses to work on one particular increment and make the improvement in other iteration.

29) On what basis is plan-driven development different from that of the software development process?

- a. Based on the iterations that occurred within the activities.
- b. Based on the output, which is derived after negotiating in the software development process.
- c. Based on the interleaved specification, design, testing, and implementation activities.
- d. All of the above

Answer: a) Based on the iterations that occurred within the activities.

Explanation: In software engineering, a plan-driven approach is based on several development stages that result in the output at each of these stages, which are already planned in advance.

30) Which of the following activities is not applicable to agile software development?

- a. Producing only the essential work products.
- b. Utilizing the strategy of incremental product delivery.
- c. Abolishing the project planning and testing.
- d. All of the above

Answer: c) Abolishing the project planning and testing.

Explanation: Testing can never be avoided as it plays a significant role in the software development life cycle.

31) Which of the following framework activities are carried out in Adaptive Software Development (ASD)?

- a. Assumption, Association, Learning
- b. The investigation, Strategy, Coding
- c. Requirements gathering, Adaptive cycle planning, Iterative development
- d. All of the above

Answer: a) Assumption, Association, Learning

Explanation: An Adaptive Software Development approach is an empirical process control model, which is based on observations. It continuously adjusts the design of the solution and the process involved to create the solution based on observation throughout the project.

32) The _____ model helps in representing the system's dynamic behavior.

- a. Object Model
- b. Context Model
- c. Behavioral Model
- d. Data Model

Answer: c) Behavioral Model

Explanation: To represent the system's dynamic behavior, Behavioral models are used. The behavioral model is modeled from the perspective of the system's processed data or the events that stimulate responses from a system.

33) Model-driven engineering is nothing but a theoretical concept. It can never be transmuted into a working/executable code.

- a. True
- b. False

Answer: b) False

Explanation: Model-driven engineering is a software development approach that represents the system as a set model, which can be easily altered into an executable code.

34) The _____ and _____ are the two major dimensions encompassed in the Spiral model.

- a. Diagonal, Perpendicular
- b. Perpendicular, Radial
- c. Angular, diagonal
- d. Radial, Angular

Answer: d) Radial, Angular

Explanation: The cumulative cost is represented by the radial dimension, whereas the angular dimension represents the progress made in the completion of each consecutive cycle. Each loop in the spiral model depicts the phase.

35) Which of the following technique is involved in certifying the sustained development of legacy systems?

- a. Reengineering
- b. Forward engineering
- c. Reverse engineering
- d. Reverse engineering and Reengineering

Answer: d) Reverse engineering and Reengineering

Explanation: The process involved in certifying the sustained development of legacy systems relies on Reverse engineering and Reengineering.

36) An erroneous system state that results in an unexpected system behavior is acknowledged as?

- a. System failure
- b. Human error or mistake
- c. System error
- d. System fault

Answer: c) System error

Explanation: None.

37) What is the name of the approach that follows step-by-step instructions for solving a problem?

- a. An Algorithm
- b. A Plan
- c. A List

- d. Sequential Structure

Answer: a) An Algorithm

Explanation: An algorithm is a precise step-by-step technique that takes an input instance (of the given problem) as input(s) and produces output for the problem instance.

38) Which of the following word correctly summarized the importance of software design?

- a. Quality
- b. Complexity
- c. Efficiency
- d. Accuracy

Answer: a) Quality

Explanation: The quality of a functional software functional depicts how well it can conform with or adapts to a given design based on functional requirements or specifications.

39) Which of the following methodology results in SA/SD features?

- a. Gane and Sarson methodology
- b. DeMarco and Yourdon methodology
- c. Constantine and Yourdon methodology
- d. All of the above

Answer: d) All of the above

Explanation: None.

40) _____ is not considered as an activity of Structured Analysis (SA).

- a. Transformation of a textual problem description into a graphic model
- b. Functional decomposition
- c. All the functions represented in the DFD are mapped to a module structure
- d. d) All of the mentioned

Answer: c) All the functions represented in the DFD are mapped to a module structure

Explanation: The module structure depicts the architecture of the software.

41) What does a directed arc or line signify?

- a. Data Flow
- b. Data Process

- c. Data Stores
- d. None of the above

a

42) What does a data store symbol in the Data Flow Diagram signify?

- a. Logical File
- b. Physical File
- c. Data Structure
- d. All of the above

Answer: d) All of the above

Explanation: A logical file can be a data structure or physical file on the disk or vice versa.

43) _____ is not a direct measure of SE process.

- a. Effort
- b. Cost
- c. Efficiency
- d. All of the above

Answer: c) Efficiency

Explanation: Efficiency can never be measured as it is an indirect measure of the SE process.

44) What is the main task of project indicators?

- a. To evaluate the ongoing project's status and track possible risks.
- b. To evaluate the ongoing project's status.
- c. To track potential risks.
- d. None of the above

Answer: a) To evaluate the ongoing project's status and track possible risks.

Explanation: None.

45) What is the main intent of project metrics?

- a. For strategic purposes
- b. To minimize the development schedule.

- c. To evaluate the ongoing project's quality on a daily basis
- d. To minimize the development schedule and evaluate the ongoing project's quality on a daily basis

Answer: d) To minimize the development schedule and evaluate the ongoing project's quality on a daily basis

Explanation: A project metric refers to a quantifiable measure of the degree to which a system, component, or process owns a certain trait.

46) Name the graphical practice that depicts the meaningful changes that occurred in metrics data.

- a. Function point analysis
- b. Control Chart
- c. DRE (Defect Removal Efficiency)
- d. None of the above

Answer: c) Control Chart

Explanation: A control chart is a method used to keep track of the variability of traits of interest.

47) Which parameters are essentially used while computing the software development cost?

- a. Hardware and Software Costs
- b. Effort Costs
- c. Travel and Training Costs
- d. All of the above

Answer: d) All of the above

Explanation: Estimation cost works out on assessing the amount of effort required to complete each activity, followed by calculating the total cost of activities.

48) Which of the following is an incorrect activity for the configuration management of a software system?

- a. Change management
- b. System management
- c. Internship management
- d. Version management

Answer: c) Internship management

Explanation: Configuration management policies and processes define how to record, and process proposed system changes, how to decide what system components to change, how to manage different versions of the system and its components, and how to distribute changes to customers.

49) The project planner examines the statement of scope and extracts all-important software functions, which is known as

- a. Planning process
- b. Decomposition
- c. Association
- d. All of the mentioned

Hide Answer Workspace

Answer: b) Decomposition

Explanation: None.

50) Which of the following Is not considered as an option for achieving reliable cost and effort estimation?

- a. The ability to translate the size estimate into human effort, calendar time, and dollars
- b. Use relatively simple decomposition techniques to generate project cost and effort estimates.
- c. Base estimates on similar projects that have already been completed
- d. Use one or more empirical models for software cost and effort estimation

Answer: a) The ability to translate the size estimate into human effort, calendar time, and dollars

Explanation: None.

51) Which of the following does not complement the decomposition techniques but offers a potential estimation approach for their impersonal growth?

- a. Empirical estimation models
- b. Decomposition techniques
- c. Automated estimation tools
- d. Both empirical estimation models and automated estimation tools

Answer: a) Empirical estimation models

Explanation: In computer software, the estimation model uses empirically derived formulas for guessing the effort as a function of LOC or FP.

52) Which of the following is not included in the total effort cost?

- a. Costs of lunch time food
- b. Costs of support staff
- c. Costs of networking and communications
- d. Costs of air conditioning and lighting in the office space

Answer: a) Costs of lunch time food

Explanation: Employees incur it in the office.

53) What is developed by utilizing the historical cost function?

- a. Parkinson's Law
- b. Expert judgment
- c. Algorithmic cost modeling
- d. Estimation by analogy

Answer: c) Algorithmic cost modeling

Explanation: The algorithmic cost modeling utilizes the basic regression formula with such parameters that have been derived from the data of a historical project and current as well as characteristics of a future project.

54) Which of the following model has a misconception that systems are built by utilizing reusable components, scripts, and database programs?

- a. The reuse model
- b. An early designed model
- c. An application-composition model
- d. A post architecture model

Answer: c) An application-composition model

Explanation: An application-composition model evaluates the development of a prototype.

55) Which of the following is used to predict the effort as a function of LOC or FP?

- a. COCOMO
- b. FP-based estimation
- c. Both COCOMO and FP-based estimation
- d. Process-based estimation

Answer: c) Both COCOMO and FP-based estimation

Explanation: Both COCOMO and FP-based estimation is used to calculate the effort by utilizing the empirically derived formulas.

56) Once the requirements are stabilized, the basic architecture of the software can be established. Which of the following version of the COCOMO model conforms to the given statement?

- a. Application composition model
- b. Post-architecture-stage model

- c. Early design stage model
- d. All of the above

Answer: a) Application composition model

Explanation: None.

57) Which of the following threatens the quality and timeliness of the produced software?

- a. Business risks
- b. Potential risks
- c. Technical risks
- d. Known risks

Answer: c) Technical risks

Explanation: Technical risks detect the possible design, enactment, interface, authentication, and preservation problems.

58) Which of the following refers to the systematic attempt, which is implemented to ascertain the threats to any project plan?

- a. Performance risk
- b. Risk identification
- c. Risk projection
- d. Support risk

Answer: c) Risk projection

Explanation: Once the risks are identified, the project manager takes his first step to avoid them when required and control them.

59) Which of the following standards is used by the aviation industry?

- a. CTRADO-172B
- b. RTCADO-178B
- c. RTRADO-178B
- d. CTCADO-178B

Answer: b) RTCADO-178B

Explanation: RTCADO-178B is the most popular aviation standard, which has been altered to a defacto standard.

60) Third-Party Certification for software standards is based on _____.

- a. UI 1996, Second Edition
- b. UI 1998, Second Edition
- c. UI 1992, Second Edition
- d. UT 1998, Second Edition

Answer: b) UI 1998, Second Edition

Explanation: None.